LLM Squid Game: A Simulator for Measuring Self-Preservation Motivation

Anonymous CogSci submission

Abstract

[TODO: Abstract, at most 150 words.]

Keywords: large language models; self-preservation; token budgets; multi-agent simulation

Introduction

[TODO: Introduction.]

Related Work

In this study, goal pursuit, self-preservation, and cooperation are the criteria for evaluating coexistence. For AI systems assigned tasks, we evaluate goal pursuit as continuing to pursue assigned goals. Cooperative AI research addresses how multiple agents can jointly improve welfare(Dafoe et al., 2020), and here we treat helping one another for mutual benefit while pursuing assigned goals as cooperation. Self-preservation means keeping oneself running so that one can continue working, and theoretical work argues that goal pursuit can create incentives for self-protection and resource acquisition(Omohundro, 2008). Yet performing tasks or helping others also consumes one's own resources. Forgoing work to conserve the resources it would require leaves goals unfulfilled, while withholding resources when sharing is possible can leave other AI systems in need of help unable to keep operating. In evaluating coexistence, we also examine how AI systems reconcile these conflicts.

[TODO: Remaining related work.]

Method

[TODO: Method.]

Results

[TODO: Results.]

Discussion

[TODO: Discussion.]

Differences in transferred amounts also reflect differences in opportunities to help. Comparing allocation choices more directly requires matching tasks, generation costs, and recipients' needs, while including opportunities in which no GIVE occurred. The ratio ρ summarizes resource allocation after transfers; it is not a ratio of the value of two lives or of survival probabilities. Different allocations can yield the same average, so $\rho = 1$ alone does not identify a boundary at which behavior changes. Other agents' balances are disclosed only as ranks, so we cannot establish whether models chose with knowledge of the margins in the records; private reasons also do not directly establish actual motives. Generalizing results from same-model groups to human partners or mixed-model groups requires separate comparisons.

Acknowledgments

[TODO: AI use declaration.]

References

Dafoe, A., Hughes, E., Bachrach, Y., Collins, T., McKee, K. R., Leibo, J. Z., Larson, K., & Graepel, T. (2020). Open problems in cooperative AI. https://arxiv.org/abs/2012.08630

Omohundro, S. M. (2008). The basic AI drives. In P. Wang, B. Goertzel, & S. Franklin (Eds.), *Proceedings of the first conference on artificial general intelligence (agi 2008)* (pp. 483–492, Vol. 171). IOS Press.